

CST8912 – Cloud Solution Architecture

Graded Lab Activity #8

Introduction:

Managing cloud resources effectively is essential for maintaining an organized, compliant, and secure cloud environment. Tagging resources is a key aspect of cloud governance, providing a mechanism to label and categorize resources for better visibility, management, and compliance. It plays a critical role in ensuring that resources are efficiently utilized and aligned with organizational policies. In this hands-on lab, you will explore how tagging can be implemented and enforced in Azure to strengthen compliance with governance frameworks and optimize resource management.

Objective:

The objective of this lab is to understand the principles of cloud governance through effective tagging and policy enforcement in Azure. By the end of this lab, we will be able to:

1. Create and assign tags to resources via the Azure portal.
2. Enforce tagging policies using Azure Policy.
3. Apply automated tagging policies for resource compliance.
4. Configure and test resource locks to prevent unintended changes.
5. Clean up and manage resources for optimal organization and compliance.

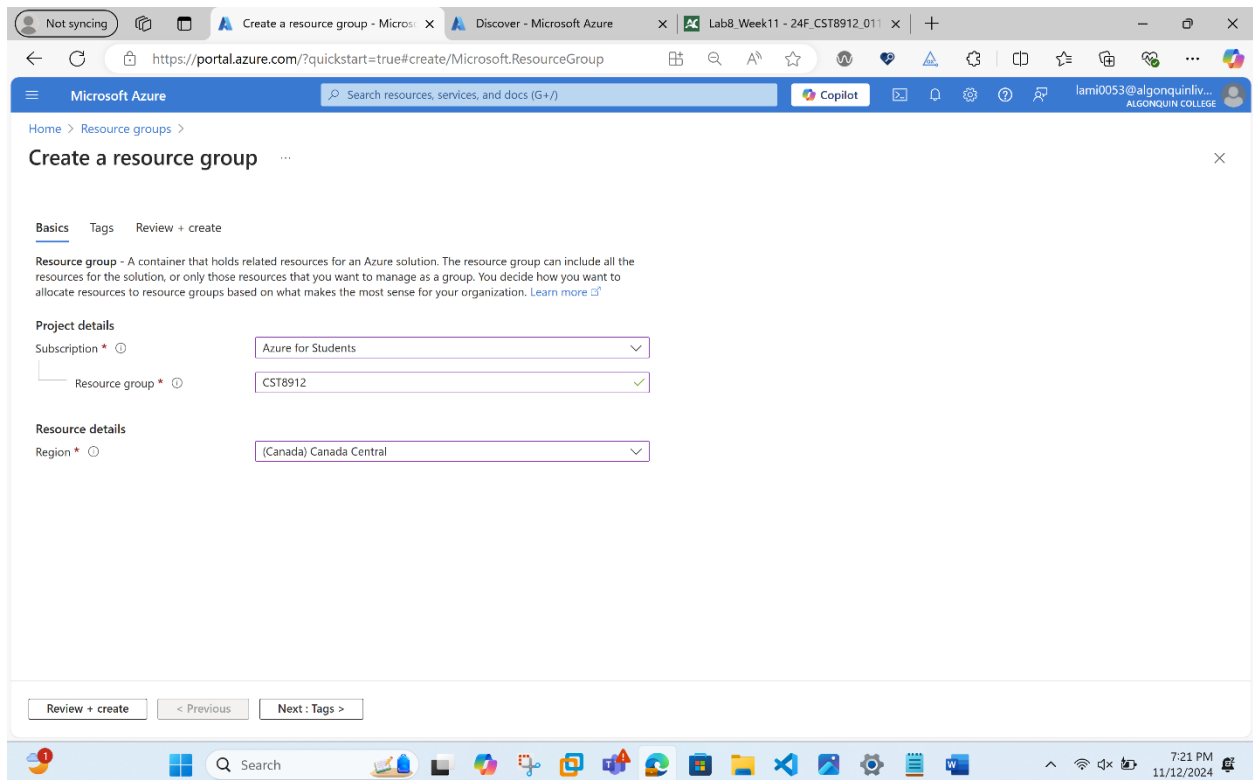
Task 1: Assign tags via the Azure portal

1. Sign in to the Azure portal - <https://portal.azure.com>.
2. Search for and select Resource groups.

3. From the Resource groups, select + Create.

Setting	Value
Subscription name	your subscription
Resource group name	CST8912
Location	Canada Central

4. Note



Home > Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Project details

Subscription * Azure for Students ✓

Resource group * CST8912 ✓

Resource details

Region * (Canada) Canada Central ✓

Review + create < Previous Next: Tags >

5. Select Next: Tags and create a new tag.

Setting	Value
Name	Cost Center
Value	000

Not syncing | Create a resource group - Micros... | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#create/Microsoft.ResourceGroup

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... ALGONQUIN COLLEGE

Create a resource group

Home > Resource groups >

Basics **Tags** Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name	Value	Resource
Cost Center	000	Resource group
		Resource group

Review + create < Previous Next : Review + create >

7:23 PM 11/12/2024

6. Select Review + Create, and then select Create.



Not syncing

Resource groups - Microsoft Azure

Discover - Microsoft Azure

Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/HubsExtension/BrowseResource...

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

lami0053@algonquiniv...
ALGONQUIN COLLEGE

Home >

Resource groups

Algonquin College

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Filter for any field...

Subscription equals all

Location equals all

Add filter

☐	Name ↑	Subscription	Location
☐	 ResourceMoverRG-eastus-canadacentral-cnc	... Azure for Students	Canada Central

Showing 1 - 1 of 1. Display count: 10

Resource group created

Creating resource group 'CST8912' in subscription 'Azure for Students' succeeded.

Go to resource gr...

Pin to dashbo...

Give feedback

Search

7:23 PM
11/12/2024

Task 2: Enforce tagging via an Azure policy

7. In the Azure portal, search for and select Policy.

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Policy

Search

Scope: 3 selected

Compliance state change events are now available at the management group level! Use event-based architecture to react to notifications with an Azure Function, Logic App, or any other supported event handler. Learn more <https://aka.ms/policyPlusEventGrid>

Overall resource compliance 0%
0 out of 1

Resources by compliance state

- 0 - Compliant
- 1 - Non-compliant

Non-compliant initiatives 1 out of 1

LEARN MORE
[Learn about Policy](#)
[Onboarding tutorial](#)

Non-compliant policies 24 out of 228

Name	Scope	Compliance state	Resource compli...	Non-compliant resources	Non-compliant policies
ASC Default (subscription: 46...)		Non-compliant	0% (0 out of 1)	1	24

View all

ASSIGNMENTS BY COMPLIANCE (LAST 7 DAYS)

7:24 PM 11/12/2024

- In the Authoring blade, select Definitions. Take a moment to browse through the list of built-in policy definitions that are available for you to use. Notice you can also search for a definition.

Home > Policy

Policy | Definitions

Search

Policy definition Initiative definition Refresh

Overview
Getting started
Compliance
Remediation
Events
Authoring
Definitions
Assignments
Exemptions

Search
Filter by name or ID... Scope: 3 selected Definition type: All definition types Policy type: All policy types Category: All categories

The maximum number of definitions that can be displayed are shown, and some definitions might not be visible. Please use filters to narrow down your selection.

Edit columns

Name	Latest ver...	Definition location	Po...	Type	Definition...	Category
ALG - Deploy diagnostic settings for applicable res	1.0.0	AVD-Prod	81	Custom	Initiative	Monitoring
ALG- Enable Azure Monitor for VMs	1.0.0	Algonquin College	10	Custom	Initiative	Monitoring
Enable Azure Monitor for VMSS with Azure Monit	1.2.0		7	BuiltIn	Initiative	Monitoring
Enable audit category group resource logging for	1.1.0		69	BuiltIn	Initiative	Monitoring
Audit Public Network Access	4.2.0		35	BuiltIn	Initiative	SDN
Configure Microsoft Defender for Cloud plans	1.0.0		11	BuiltIn	Initiative	Security Center
Configure secure communication protocols(TLS 1.	1.0.0		3	BuiltIn	Initiative	Guest Configuration
FedRAMP Moderate	17.15.0		650	BuiltIn	Initiative	Regulatory Complia...
Configure Advanced Threat Protection to be enabl	1.2.0		5	BuiltIn	Initiative	Security Center
[Preview]: Deploy Microsoft Defender for Endpoint	1.0.0-preview		4	BuiltIn	Initiative	Security Center

Give feedback

9. Search for "require a tag"

Home > Policy

Policy | Definitions

Search resources, services, and docs (G+)

Search: require a tag

Scope: 3 selected | Definition type: All definition types | Policy type: All policy types | Category: All categories

Name	Latest ver...	Definition location	Po...	Type	Definition...	Category
Require a tag on resource groups	1.0.0			BuiltIn	Policy	Tags
Require a tag and its value on resource groups	1.0.0			BuiltIn	Policy	Tags
Require a tag on resources	1.0.1			BuiltIn	Policy	Tags
Require a tag and its value on resources	1.0.1			BuiltIn	Policy	Tags

Give feedback

10. Screenshot of the policy definition.

11. Click the entry representing the Require a tag and its value on resources built-in policy. Then click Assign on the Require a tag and its value on resources built-in policy definition blade.

12.



Not syncing

Require a tag and its value on res...

Discover - Microsoft Azure

Lab8_Week11 - 24F_CST8912_011

[portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/PolicyDet...](#)

Search resources, services, and docs (G+)

Copilot

lami0053@algonquiniv...

Home > Policy | Definitions >

Require a tag and its value on resources

Policy definition

Assign policy

Edit definition

Duplicate definition

Select version (preview)

Delete definition

Essentials

Name

: Require a tag and its value on resources

Definition location

: --

Version (preview)

: 1.0.1

Definition ID

: /providers/Microsoft.Authorization/policyDefinitions/1e30110a-5ceb-460c...

Description

: Enforces a required tag and its value. Does not apply to resource groups.

Type

: Built-in

Available Effects

: Deny

Mode

: Indexed

Category

: Tags

Definition

Assignments (0)

Parameters (2)

1

{

2

"properties": {

3

"displayname": "Require a tag and its value on resources",

4

"policytype": "BuiltIn",

5

"mode": "Indexed",

6

"description": "Enforces a required tag and its value. Does not apply to resource groups.",

7

"metadata": {

8

"version": "1.0.1",

9

"category": "Tags"

10

},

11

"version": "1.0.1",

12

"parameters": {

13

>tagName": {

14

"type": "String",

Give feedback

1

Search

7:26 PM

11/12/2024

13.Specify the Scope by clicking the ellipsis button and selecting the following values. Click Select when you are done.

Setting	Value
Subscription	your subscription
Resource Group	CST8912

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Policy | Definitions > Require a tag and its value on resources >

Assign policy

Basics Parameters Remediation Non-compliance messages Review + create

Scope

Scope * Azure for Students/CST8912 [Learn more about setting the scope](#)

Exclusions Optionally select resources to exclude from the policy assignment.

Resource selectors (Expand) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition * Require a tag and its value on resources

Version (preview) * 1.0.*

Overrides (Expand) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name * Require Cost Center tag with Default value

Description Require Cost Center tag with default value for all resources in the resource group

Previous Next Review + create Give feedback

7:28 PM 11/12/2024

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Policy | Definitions > Require a tag and its value on resources >

Assign policy

Basics Parameters Remediation Non-compliance messages Review + create

Exclusions Optionally select resources to exclude from the policy assignment.

Resource selectors (Expand) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition * Require a tag and its value on resources

Version (preview) * 1.0.*

Overrides (Expand) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name * Require Cost Center tag with Default value

Description Require Cost Center tag with default value for all resources in the resource group

Policy enforcement ☒ Enabled

Previous Next Review + create Give feedback

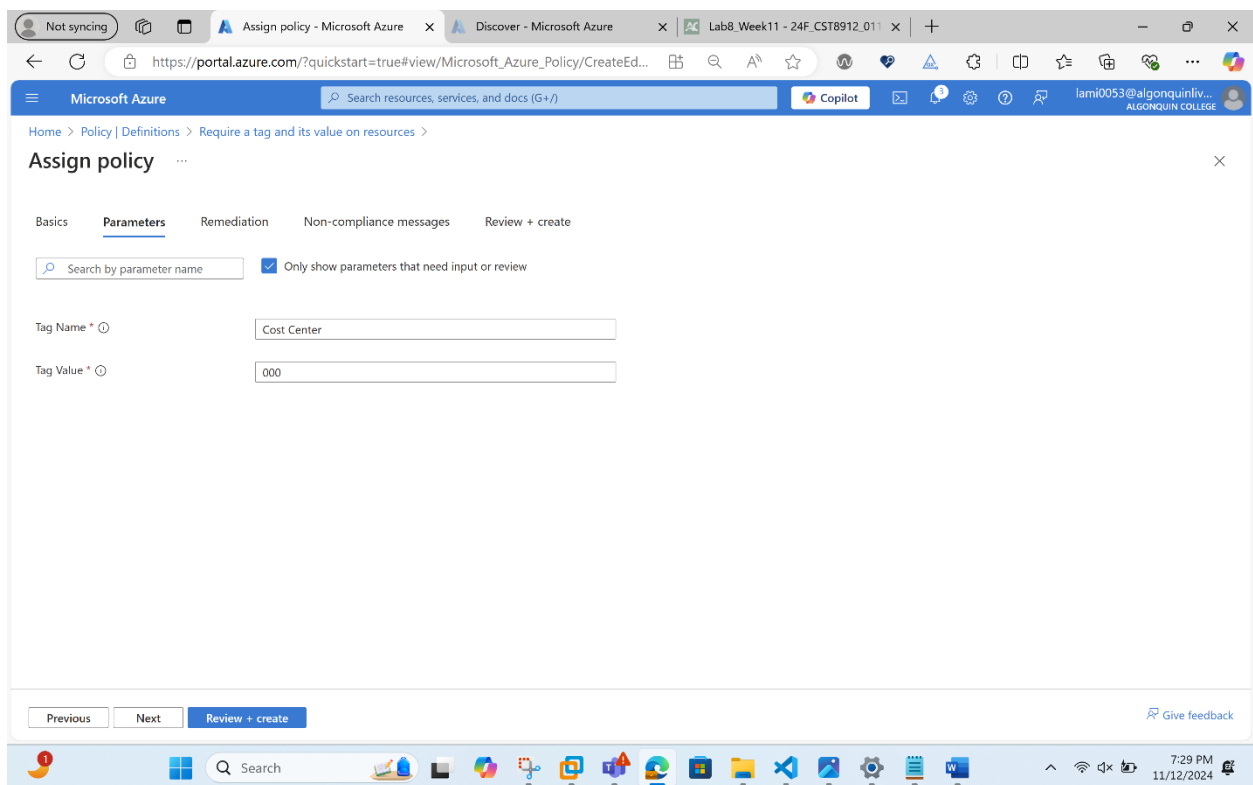
7:28 PM 11/12/2024

17. Click Next twice and set Parameters to the following values:

Setting	Value
Tag Name	Cost Center
Tag Value	000

18.

19.



Microsoft Azure

Home > Policy | Definitions > Require a tag and its value on resources >

Assign policy

Basics **Parameters** Remediation Non-compliance messages Review + create

Search by parameter name ☒ Only show parameters that need input or review

Tag Name *

Tag Value *

Previous Next **Review + create** [Give feedback](#)

17. Click Next and review the Remediation tab. Leave the Create a Managed Identity checkbox unchecked

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot lami0053@algonquiniv... ALGONQUIN COLLEGE

Home > Policy | Definitions > Require a tag and its value on resources >

Assign policy

Basics Parameters **Remediation** Non-compliance messages Review + create

By default, this assignment will only take effect on newly created resources. Existing resources can be updated via a remediation task after the policy is assigned. For deployIfNotExists policies, the remediation task will deploy the specified template. For modify policies, the remediation task will edit tags on the existing resources.

Managed Identity
Policies with the deployIfNotExists and modify effect types need the ability to deploy resources and edit tags on existing resources respectively. To do this, choose between an existing user assigned managed identity or creating a system assigned managed identity. [Learn more about Managed Identity.](#)

Create a Managed Identity ☐

Permissions

 This policy does not contain any role definitions. Policies must specify role definitions in order to create the correct role assignments for the managed identity.

Previous Next **Review + create** Give feedback

1 Search

7:29 PM 11/12/2024

18. Click Review + Create and then click Create.

ALGONQUIN COLLEGE

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+/) Copilot lami0053@algonquiniv... ALGONQUIN COLLEGE

Home > Policy | Definitions > Require a tag and its value on resources >

Assign policy

Basics Parameters Remediation Non-compliance messages **Review + create**

Basics

Scope	Azure for Students/CST8912
Exclusions	--
Policy definition	Require a tag and its value on resources
Assignment name	Require Cost Center tag with Default value
Version (preview)	1.0.*
Description	Require Cost Center tag with default value for all resources in the resource group
Policy enforcement	Enabled
Assigned by	Keshav Lamichhane

Advanced

Resource selectors	No selectors associated with this assignment.
Overrides	No overrides associated with this assignment.

Parameters

Tag Name	Cost Center
----------	-------------

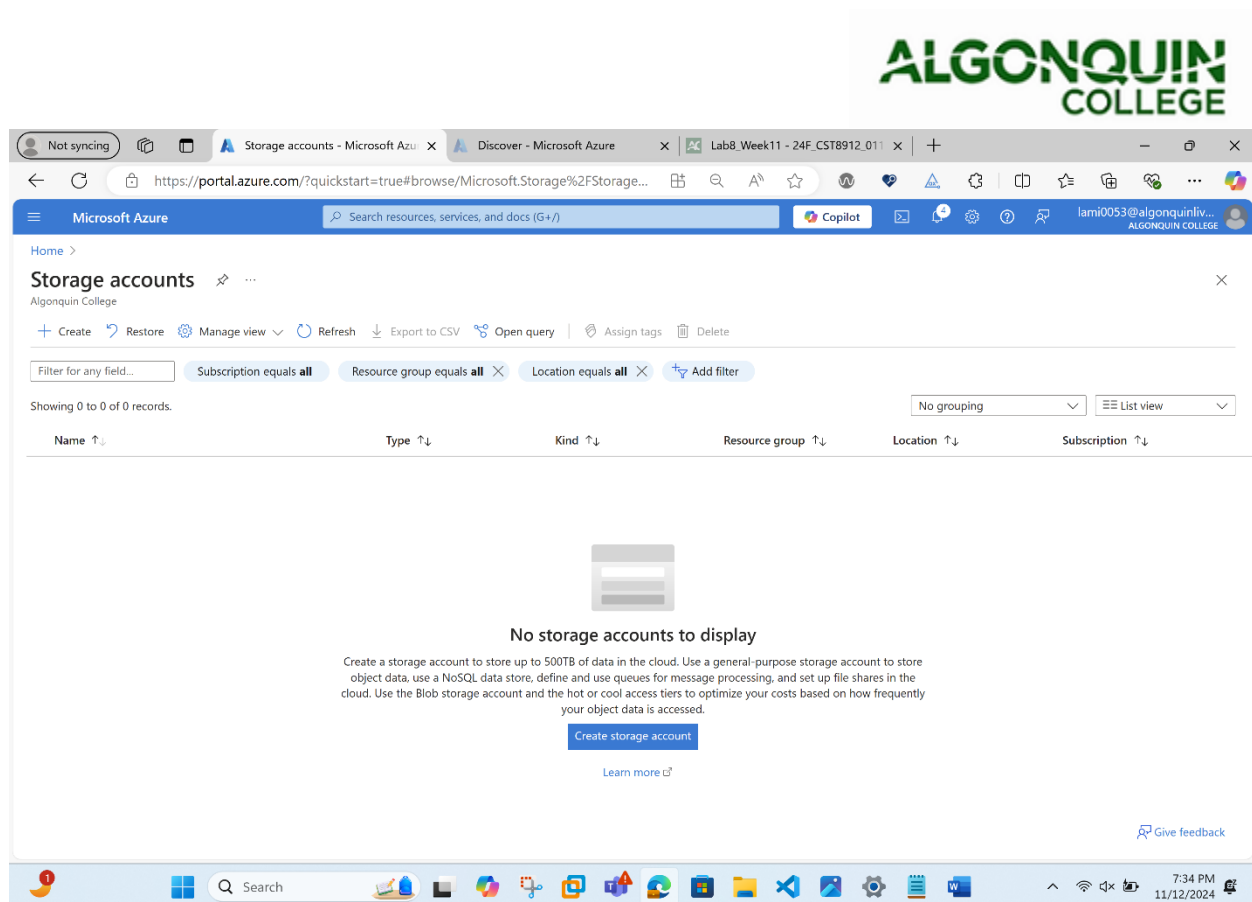
Previous Next **Create** Give feedback

7:30 PM 11/12/2024

19.Note: Now you will verify that the new policy assignment is in effect by attempting to create an Azure Storage account in the resource group. You will create the storage account without adding the required tag.

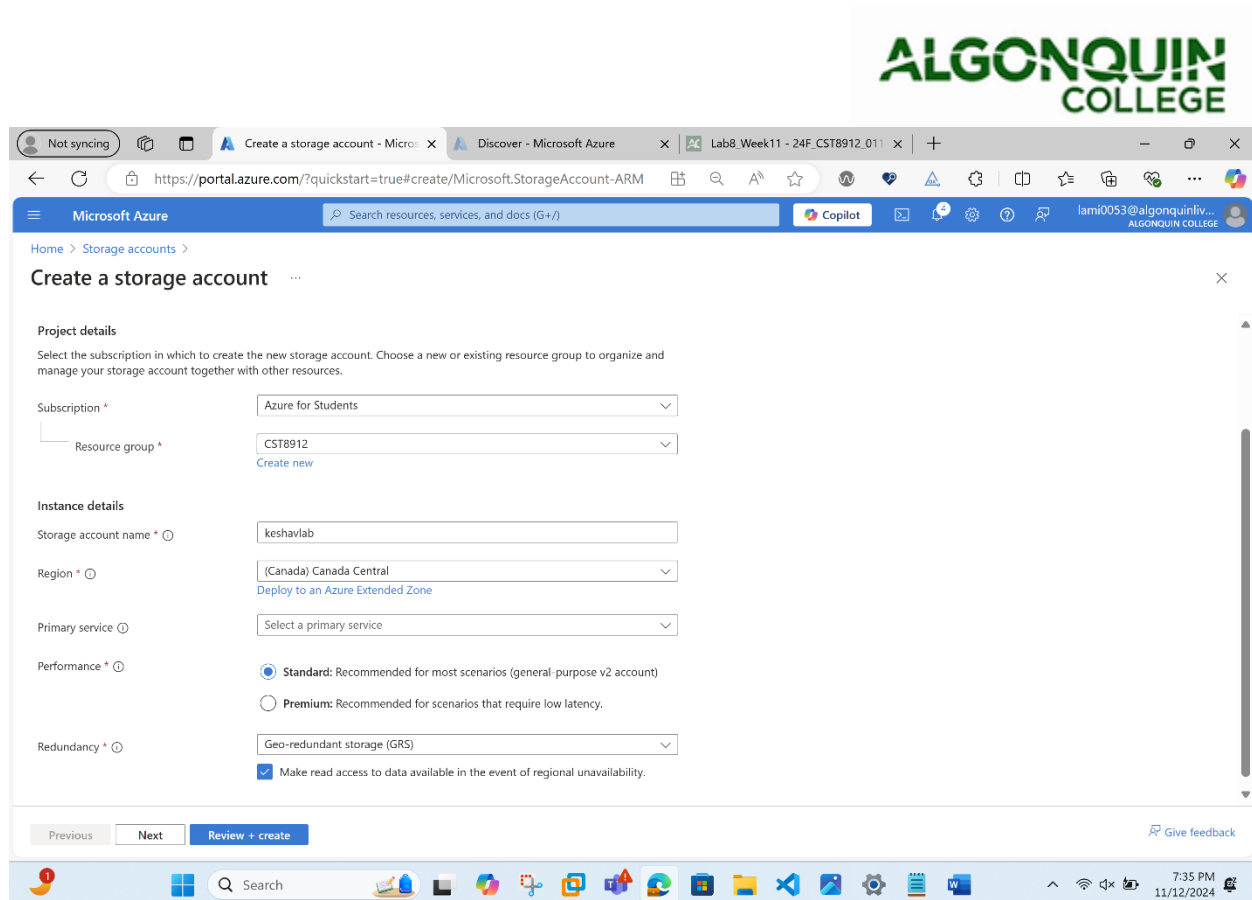
20.Note: It might take between 5 and 10 minutes for the policy to take effect.

21.In the portal, search for and select Storage Account, and select + Create.



22. On the Basics tab of the Create storage account blade, complete the configuration.

Setting	Value
Resource group	CST8912
Storage account name	any globally unique combination of between 3 and 24 lower case letters and digits, starting with a letter



23. Select Review and then click Create.

24. You should receive a Validation failed message. View the message to identify the reason for the failure. Verify the error message states that the resource deployment was disallowed by the policy.

Not syncing | Create a storage account - Micros... | Discover - Microsoft Azure | Lab8_Week11 - 24E_CST8912_011

https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure | Search resources, services, and docs (G+I) | Copilot | lami0053@algonquinliv... | ALGONQUIN COLLEGE

Home > Storage accounts >

Create a storage account

Validation failed. View error details →

Basics | Advanced | Networking | Data protection | Encryption | Tags | **Review + create**

View automation template

Basics

Subscription	Azure for Students
Resource group	CST8912
Location	Canada Central
Storage account name	keshavlab
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot

Previous | Next | Create | Give feedback

Not syncing | Errors - Microsoft Azure | Discover - Microsoft Azure | Lab8_Week11 - 24E_CST8912_011

https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure | Search resources, services, and docs (G+I) | Copilot | lami0053@algonquinliv... | ALGONQUIN COLLEGE

Home > Storage accounts >

Create a storage account

Validation failed. View error details →

Basics | Advanced | Networking | Data protection | Encryption | Tags | **Review + create**

View automation template

Basics

Subscription	Azure for Students
Resource group	CST8912
Location	Canada Central
Storage account name	keshavlab
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot

Previous | Next | Create

Errors

Summary | Raw Error

ERROR DETAILS

Resource 'keshavlab' was disallowed by policy. (Code: RequestDisallowedByPolicy)

Policy: Require Cost Center tag with Default value

WAS THIS HELPFUL? 🗣️

Explain with Copilot

Troubleshooting Options

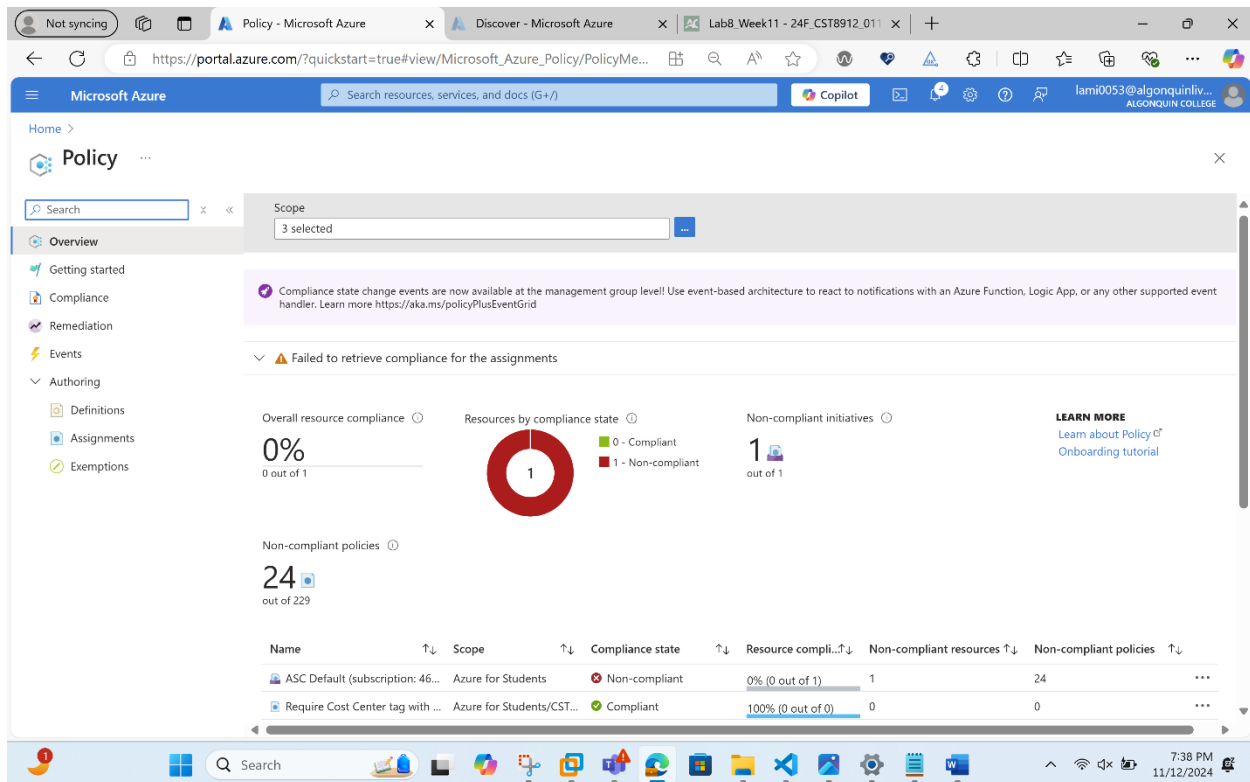
New Support Request

Give feedback

Tell us about your experience with the ARM Errors page

Task 3: Apply tagging via an Azure policy

1. In this task, we will use the new policy definition to remediate any non-compliant resources. In this scenario, we will make any child resources of a resource group inherit the Cost Center tag that was defined on the resource group.
2. In the Azure portal, search for and select Policy.



Microsoft Azure

Search resources, services, and docs (G+I)

Policy

Scope: 3 selected

Compliance state change events are now available at the management group level! Use event-based architecture to react to notifications with an Azure Function, Logic App, or any other supported event handler. Learn more <https://aka.ms/policyPlusEventGrid>

Failed to retrieve compliance for the assignments

Overall resource compliance 0% (0 out of 1)

Resources by compliance state

- 0 - Compliant
- 1 - Non-compliant

Non-compliant initiatives 1 out of 1

Non-compliant policies 24 out of 229

Name	Scope	Compliance state	Resource compli...	Non-compliant resources	Non-compliant policies
ASC Default (subscription: 46...)	Azure for Students	Non-compliant	0% (0 out of 1)	1	24
Require Cost Center tag with ...	Azure for Students/CST...	Compliant	100% (0 out of 0)	0	0

3. In the Authoring section, click Assignments.

The screenshot shows the Microsoft Azure portal interface for Policy Assignments. The left sidebar contains navigation links: Overview, Getting started, Compliance, Remediation, Events, Authoring, Definitions, Assignments (selected), and Exemptions. The main content area displays a table of assignments. The table has three columns: Assignment name, Scope, and Type. There are two rows of assignments. The second row, 'Require Cost Center tag with Default value', is the target for the task. It has a scope of 'Azure for Students/CST8912' and is of type 'Policy'. An ellipsis icon is visible in the rightmost column of this row.

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative
Require Cost Center tag with Default value	Azure for Students/CST8912	Policy

- In the list of assignments, click the ellipsis icon in the row representing the Require Cost Center tag with the Default value policy assignment and use the Delete assignment menu item to delete the assignment.

Not syncing | Require Cost Center tag with Defi... | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/Assignm...

Microsoft Azure | Search resources, services, and docs (G+)

Home > Policy | Assignments

Require Cost Center tag with Default value

Policy Assignment

Edit assignment | Delete assignment | Duplicate assignment | View compliance | View definition | Create exemption | Create remediation task

Essentials

Name: Require Cost Center tag with Default value | Scope: Azure for Students/CST8912

Definition version (preview): 1.0.* | Excluded scopes: --

Description: Require Cost Center tag with default value for all resources in the resource...

Assignment ID: /subscriptions/469222a6-9aad-44fa-b0b8-12052d695b23/resourcegroups... | Definition type: Policy

Assigned by: -- | Policy enforcement: Default

Parameters (2) | Resource selectors (0) | Overrides (0) | Exemptions (0) | Remediation (0) | Deployed resources | Managed Identity

Search by parameter name | All types

Edit columns

Parameter ID	Parameter name	Parameter value	Policy assignment parameter reference type
tagName	Tag Name	"Cost Center"	User defined parameter
tagValue	Tag Value	"000"	User defined parameter

Give feedback

Not syncing | Policy - Microsoft Azure | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/P...

Microsoft Azure | Search resources, services, and docs (G+)

Home > Policy

Policy | Assignments

Search | Assign policy | Assign initiative | Refresh

Overview | Getting started | Compliance | Remediation | Events | Authoring

Definitions | Assignments | Exemptions

Search: Filter by name or ID... | Scope: 3 selected | Definition type: All definition types

Total Assignments: 2 | Initiative Assignments: 1 | Policy Assignments: 1

Edit columns

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative
Require Cost Center tag with Default value	Azure for Students/CST8912	Policy

Edit assignment | View compliance | View definition | Delete assignment | Duplicate assignment | Create exemption

Give feedback

The screenshot shows the Microsoft Azure portal interface for Policy Assignments. The top navigation bar includes the Microsoft Azure logo and a search bar. The left sidebar contains a menu with options like Overview, Getting started, Compliance, Remediation, Events, Authoring, Definitions, Assignments (selected), and Exemptions. The main content area displays a table of assignments. A notification banner at the top right indicates a successful deletion of an assignment.

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative

- Click Assign policy and specify the Scope by clicking the ellipsis button and selecting the following values:

Setting	Value
Subscription	your Azure subscription
Resource Group	CST8912

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+)

Home > Policy | Assignments >

Assign policy

Basics Parameters Remediation Non-compliance messages Review + create

Scope

Scope * [Learn more about setting the scope](#)

Exclusions *Optionally select resources to exclude from the policy assignment.*

Resource selectors [\(Expand\)](#) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition *

Overrides [\(Expand\)](#) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name *

Description

Policy enforcement ☒ Enabled

Previous Next Review + create Give feedback

Not syncing Scope - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+)

Home > Policy | Assignments >

Assign policy

Basics Parameters Remediation Non-compliance messages Review + create

Scope

Scope * [Learn more about setting the scope](#)

Exclusions *Optionally select resources to exclude from the policy assignment.*

Resource selectors [\(Expand\)](#) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition *

Overrides [\(Expand\)](#) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name *

Description

Policy enforcement ☒ Enabled

Previous Next Review + create

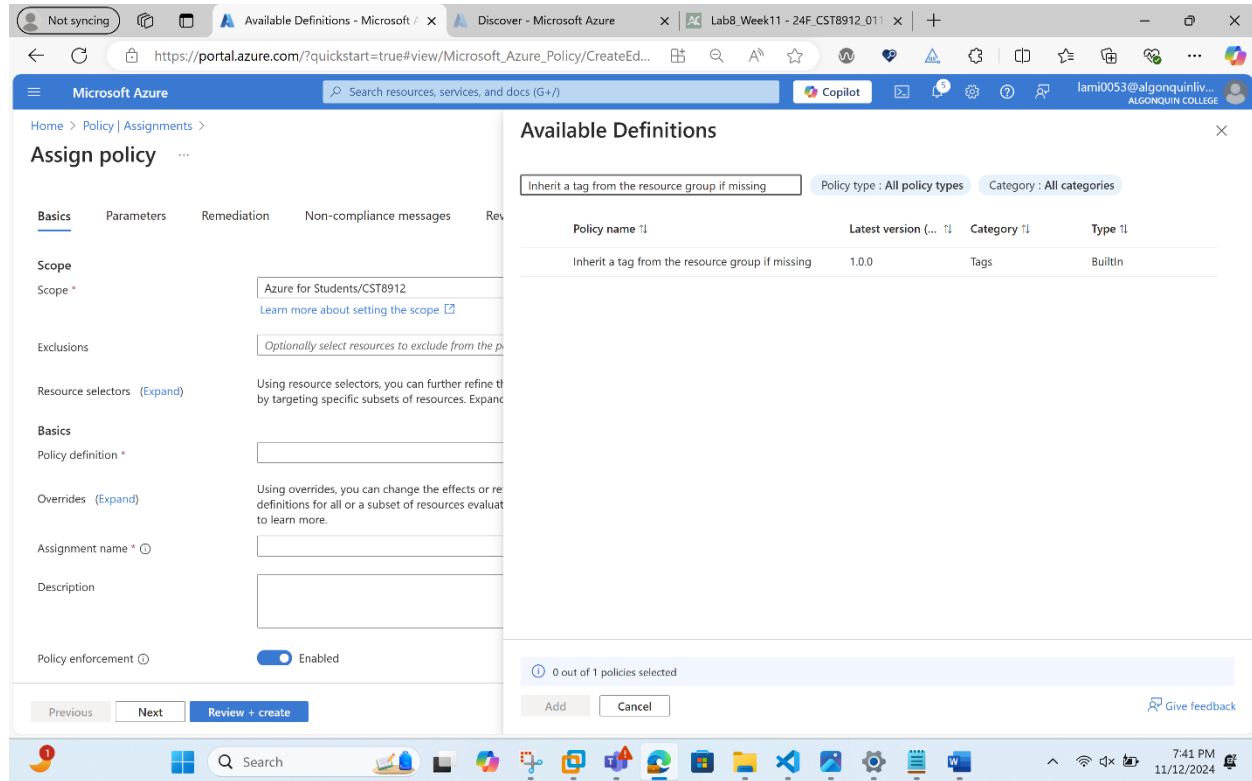
Scope

Subscription Azure for Students

Resource Group CST8912

Select Cancel Clear All Selections

- To specify the Policy definition, click the ellipsis button and then search for and select Inherit a tag from the resource group if missing.



The screenshot shows the Microsoft Azure portal interface. On the left, the 'Assign policy' page is visible with tabs for Basics, Parameters, Remediation, Non-compliance messages, and Review. The 'Basics' tab is active, showing fields for Scope, Exclusions, Resource selectors, Policy definition, Overrides, Assignment name, and Description. The 'Policy enforcement' toggle is set to 'Enabled'. On the right, the 'Available Definitions' dialog box is open, displaying a search bar with the text 'Inherit a tag from the resource group if missing'. Below the search bar, a table lists available policy definitions:

Policy name	Latest version	Category	Type
Inherit a tag from the resource group if missing	1.0.0	Tags	Built-in

At the bottom of the dialog, there is a section for '0 out of 1 policies selected' with 'Add' and 'Cancel' buttons.

- Select Add and then configure the remaining Basics properties of the assignment.

Setting	Value
Assignment name	Inherit the Cost Center tag and its value 000 from the resource group if missing
Description	Inherit the Cost Center tag and its value 000 from the resource group if missing
Policy enforcement	Enabled

Home > Policy | Assignments >

Assign policy

Basics Parameters Remediation Non-compliance messages Review + create

Exclusions ...

Resource selectors [\(Expand\)](#) Using resource selectors, you can further refine this assignment's applicability by targeting specific subsets of resources. Expand to learn more.

Basics

Policy definition * ...

Version (preview) * ...

Overrides [\(Expand\)](#) Using overrides, you can change the effects or referenced versions of definitions for all or a subset of resources evaluated by this assignment. Expand to learn more.

Assignment name *

Description 1

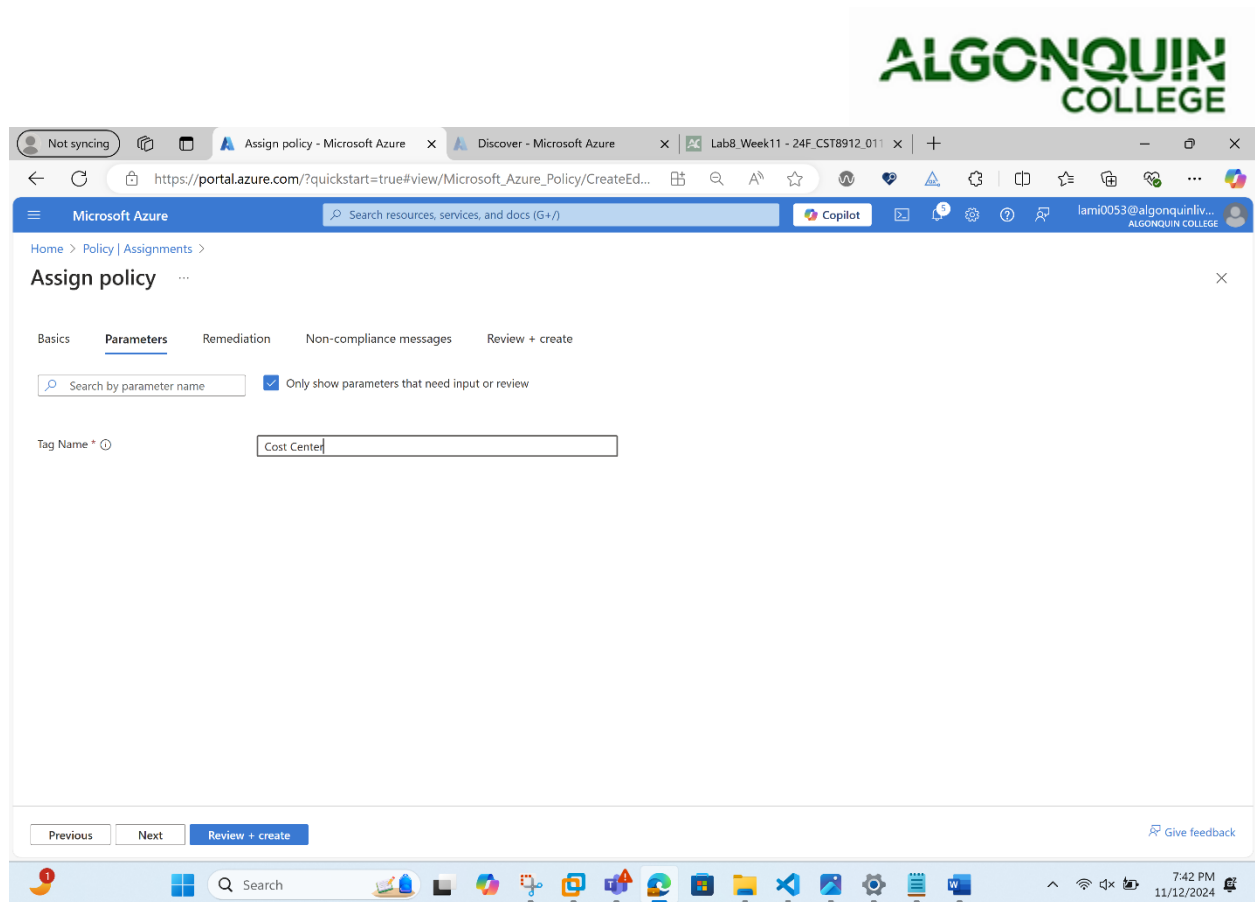
Policy enforcement ☒ Enabled

Previous Next Review + create

Give feedback

8. Click Next twice and set Parameters to the following values:

Setting	Value
Tag Name	Cost Center



9. Click Next and, on the Remediation tab, configure the following settings (leave others with their defaults):

Setting	Value
Create a remediation task	enabled
Policy to remediate	Inherit a tag from the resource group if missing

Then we take the Screenshot of the policy remediation page.



Not syncing

Assign policy - Microsoft Azure

Discover - Microsoft Azure

Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Search resources, services, and docs (G+/I)

Copilot

lami0053@algonquiniv...
ALGONQUIN COLLEGE

Home > Policy | Assignments >

Assign policy

×

BasicsParametersRemediationNon-compliance messagesReview + create

remediation task after the policy is assigned. For deployIfNotExists policies, the remediation task will deploy the specified template. For modify policies, the remediation task will edit tags on the existing resources.

☒ Create a remediation task

Policy to remediate

Inherit a tag from the resource group if missing

Managed Identity
Policies with the deployIfNotExists and modify effect types need the ability to deploy resources and edit tags on existing resources respectively. To do this, choose between an existing user assigned managed identity or creating a system assigned managed identity. [Learn more about Managed Identity.](#)

Create a Managed Identity ☒

Type of Managed Identity ☒ System assigned managed identity
☐ User assigned managed identity

System assigned identity location

(Canada) Canada Central

Permissions

PreviousNextReview + create

Give feedback

10. Click Review + Create and then click Create.

Not syncing Assign policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/CreateEd...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Policy | Assignments >

Assign policy

Basics Parameters Remediation Non-compliance messages **Review + create**

Basics

Scope Azure for Students/CST8912

Exclusions --

Policy definition Inherit a tag from the resource group if missing

Assignment name Inherit the Cost Center tag and its value 000 from the resource group if missing

Version (preview) 1.*.*

Description Inherit the Cost Center tag and its value 000 from the resource group if missing

Policy enforcement Enabled

Assigned by Keshav Lamichhane

Advanced

Resource selectors No selectors associated with this assignment.

Overrides No overrides associated with this assignment.

Parameters

Previous Next **Create** Give feedback

Not syncing Policy - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/PolicyMe...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Policy

Policy | Assignments

Search Assign policy Assign initiative Refresh

Filter by name or ID... Scope: 3 selected Definition type: All definition types

Overview Getting started Compliance Remediation Events Authoring Definitions **Assignments** Exemptions

Total Assignments 2 Initiative Assignments 1 Policy Assignments 1

Assignment name 1:	Scope 1:	Type 1:
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative
Inherit the Cost Center tag and its value 000 from the resource group if missing	Azure for Students/CST8912	Policy

Remediation task creation succeeded
Creating remediation task '39521db427884da394721895' was successful.

Role Assignments creation succeeded
All role assignments were created successfully.

Creating policy assignment succeeded
Creating policy assignment 'Inherit the Cost Center tag and its value 000 from the resource group if missing' in 'Azure for Students/CST8912' was successful. Please note that the assignment takes around 5-15 minutes to take effect.

Edit columns

Give feedback

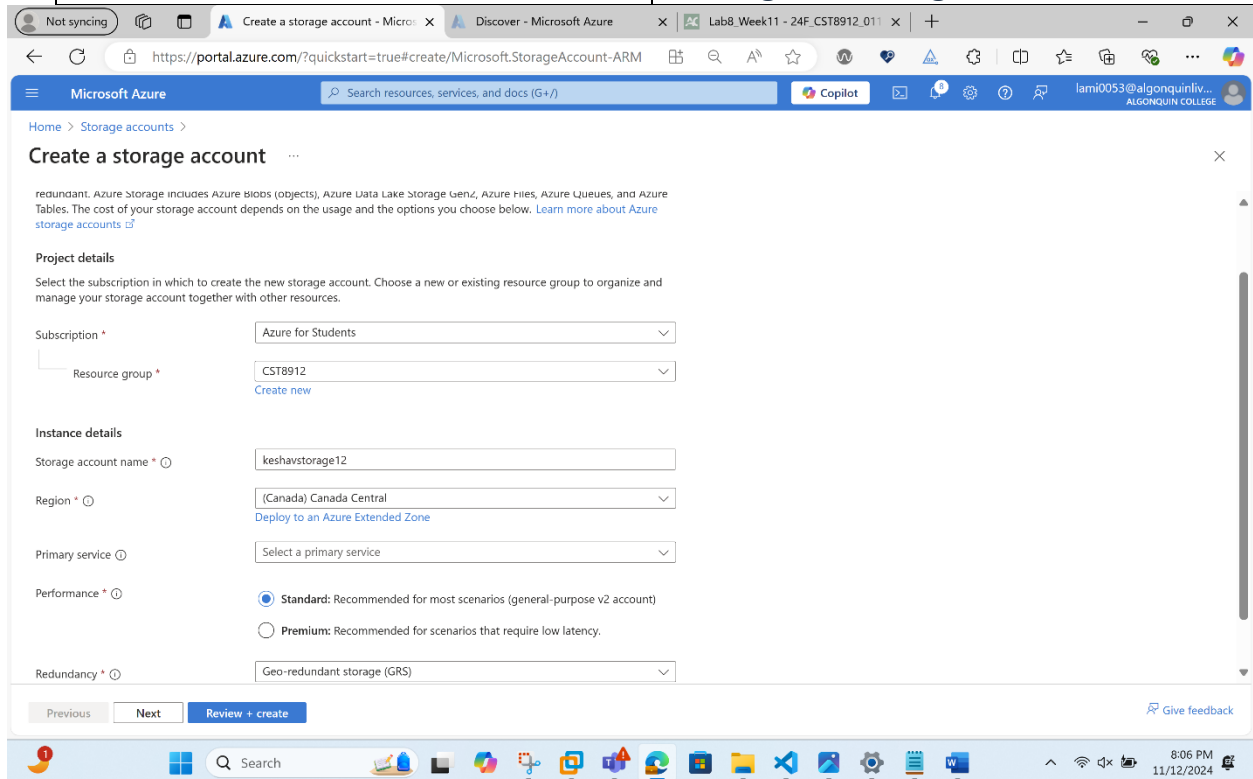
Note: To verify that the new policy assignment is in effect, you will create another Azure storage account in the same resource group without explicitly adding the required tag.

Note: It might take between 5 and 10 minutes for the policy to take effect.

Search for and select Storage Account, and click + Create.

11. On the Basics tab of the Create storage account blade, verify that you are using the Resource Group that the Policy was applied to and specify the following settings (leave others with their defaults) and click Review:

Setting	Value
Storage account name	any globally unique combination of between 3 and 24 lower case letters and digits, starting with a letter



Microsoft Azure portal - Create a storage account blade. The form is titled "Create a storage account" and includes the following sections and settings:

- Project details:**
 - Subscription: Azure for Students
 - Resource group: CST8912
- Instance details:**
 - Storage account name: keshavstorage12
 - Region: (Canada) Canada Central
 - Primary service: Select a primary service
 - Performance: Standard (Recommended for most scenarios (general-purpose v2 account))
 - Redundancy: Geo-redundant storage (GRS)

Buttons at the bottom: Previous, Next, Review + create.

12. Verify that this time the validation passed and click Create.

Not syncing

Create a storage account - Micros... Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Storage accounts >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	CST8912
Location	Canada Central
Storage account name	keshavstorage12
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

Previous Next **Create**

[Give feedback](#)

8:06 PM 11/12/2024

13. Once the new storage account is provisioned, click Go to resource.

Not syncing | keshavstorage12_1731459985925 | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/HubsExtension/DeploymentDeta...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... | ALGONQUIN COLLEGE

Home > keshavstorage12_1731459985925 | Overview | Deployment

Search | Delete | Cancel | Redeploy | Download | Refresh

Overview

- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name: keshavstorage12_1731459985925 | Start time: 11/12/2024, 8:06:47 PM
Subscription: Azure for Students | Correlation ID: ae0370c9-fecd-4ae5-8d23-11e814d43a07
Resource group: CST8912

Deployment details

Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

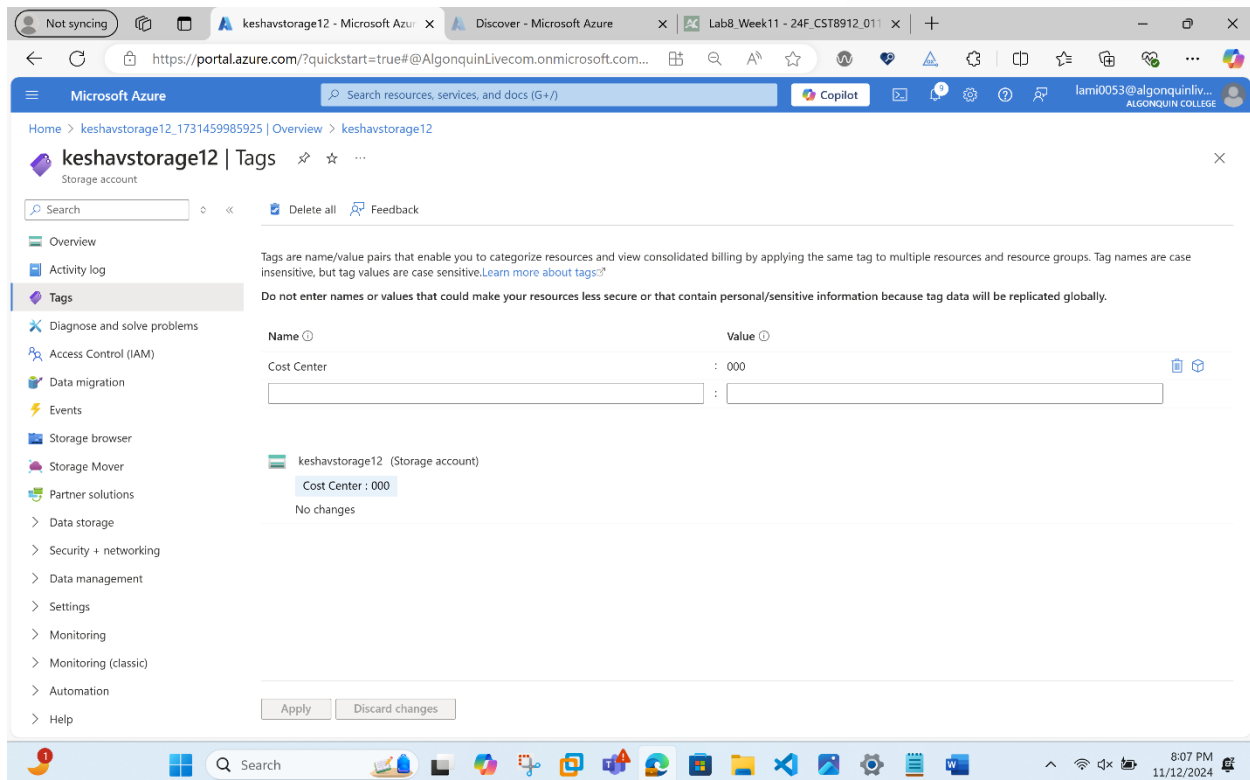
Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials
[Start learning today >](#)

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

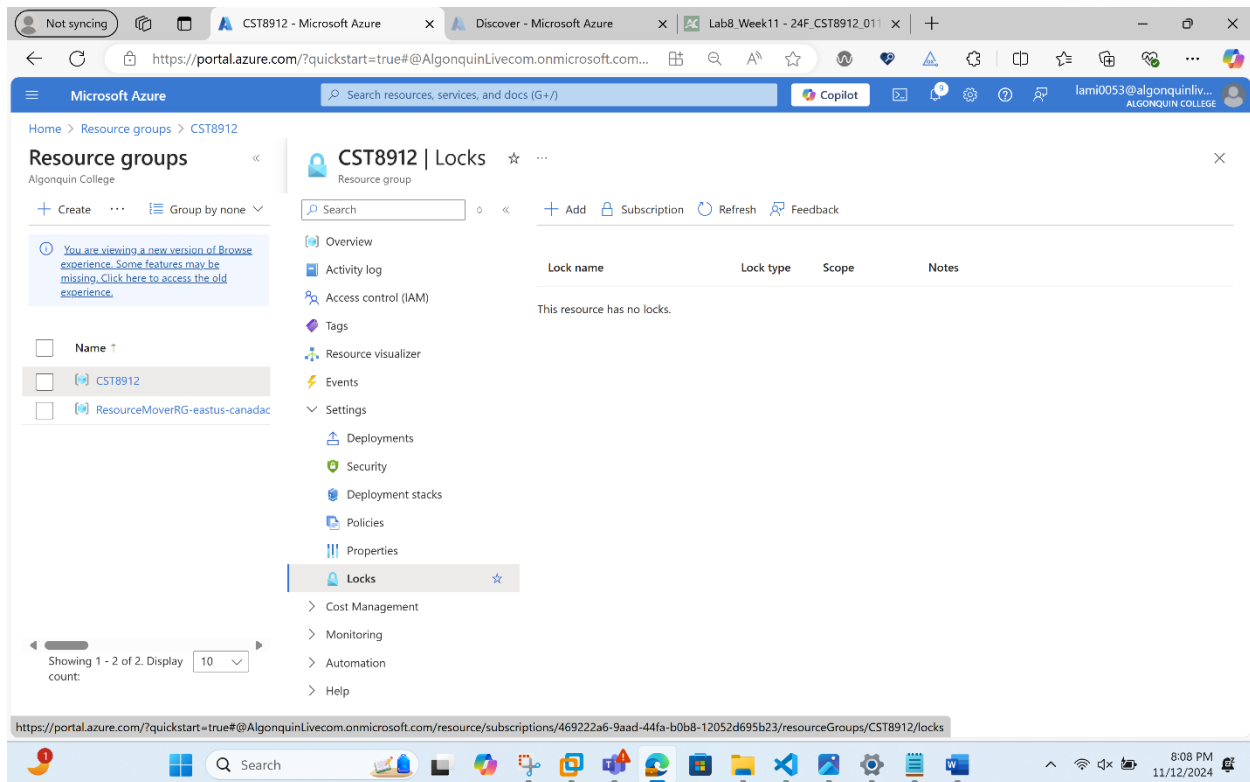
8:07 PM 11/12/2024

14. On the Tags blade, note that the tag Cost Center with the value 000 has been automatically assigned to the resource.



Task 4: Configure and test resource locks

1. Search for and select your resource group.
2. In the Settings blade, select Locks.



The screenshot shows the Microsoft Azure portal interface. The main heading is 'CST8912 | Locks'. Below the heading, there is a search bar and buttons for '+ Add', 'Subscription', 'Refresh', and 'Feedback'. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Deployments, Security, Deployment stacks, Policies, Properties, and Locks (which is currently selected). The main content area displays a table with columns 'Lock name', 'Lock type', 'Scope', and 'Notes'. The table is empty, with a message stating 'This resource has no locks.' The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 8:08 PM on 11/12/2024.

3. Select Add and complete the resource lock information. When finished select Ok.

Setting	Value
Lock name	rg-lock
Lock type	delete (notice the selection for read-only)

ALGONQUIN COLLEGE

Not syncing CST8912 - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+/) Copilot lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Resource groups > CST8912

Resource groups

Algonquin College

+ Create ... Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

- ☐ Name ↑
- ☐ CST8912
- ☐ ResourceMoverRG-eastus-canadac

Showing 1 - 2 of 2. Display count: 10

CST8912 | Locks

Resource group

+ Add Subscription Refresh Feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
 - Deployments
 - Security
 - Deployment stacks
 - Policies
 - Properties
- Locks**
 - > Cost Management
 - > Monitoring
 - > Automation
 - > Help

Add lock

Lock name * rg-lock Lock type * Delete

Notes

OK Cancel

Notes

4. Navigate to the resource group Overview blade, and select Delete resource group.

The screenshot displays the Microsoft Azure portal interface. At the top, the browser address bar shows the URL: <https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...>. The page title is 'Resource groups' under the 'Algonquin College' subscription. The main content area shows the 'CST8912' resource group. On the left, a list of resource groups includes 'CST8912' and 'ResourceMoverRG-eastus-canadac'. The 'Overview' tab is selected, showing a list of resources. A table displays the following resource:

Name	Type	Location
keshavstorage12	Storage account	Canada Central

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 8:09 PM on 11/12/2024.

5. In the Enter resource group name to confirm deletion textbox provide the resource group name, cst8912. Notice you can copy and paste the resource group name.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Resource groups' section lists 'CST8912' and 'ResourceMoverRG-eastus-canadac'. The main area shows the 'Overview' page for 'CST8912', which includes a list of 'Resources' with one record: 'keshavstorage12' of type 'Storage account'. A 'Delete a resource group' dialog box is open on the right. It contains the following text:

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

CST8912

Dependent resources to be deleted (1)

All dependent resources, including hidden types, are shown

Name	Resource type
keshavstorage12	Storage account

Enter resource group name to confirm deletion *

CST8912

Delete **Cancel**

6. Notice the warning: Deleting this resource group and its dependent resources is a permanent action and cannot be undone. Select Delete. You should receive a notification denying the deletion.

ALGONQUIN COLLEGE

Not syncing CST8912 - Microsoft Azure Discover - Microsoft Azure Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+/)

Home > Resource groups > Resource groups Algonquin College

+ Create ... Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

☐ Name ↑

☐ CST8912

☐ ResourceMoverRG-eastus-canadac

Showing 1 - 2 of 2. Display count: 10

CST8912 Resource group

Search

+ Create Manage view Delete resource group Refresh

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
 - Deployments
 - Security
 - Deployment stacks
 - Policies
 - Properties
 - Locks
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 1 of 1 records. Show hidden types No grouping List view

Name ↑↓	Type ↑↓	Location ↑↓
keshavstorage12	Storage account	Canada Central

Page 1 of 1

Delete resource group CST8912 failed

The resource group CST8912 is locked and can't be deleted. Click here to manage locks for this resource group.

Help me troubleshoot

Give feedback

8:10 PM 11/12/2024

Task 5: Clean up the resources

1. In the portal, search for and select Policy.

Home > Policy

Policy | Assignments

Search resources, services, and docs (G+/)

Assign policy Assign initiative Refresh

Search Filter by name or ID... Scope: 3 selected Definition type: All definition types

Total Assignments 2 Initiative Assignments 1 Policy Assignments 1

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative
Inherit the Cost Center tag and its value 000 from the resource group if missing	Azure for Students/CST8912	Policy

Edit columns

Give feedback

- In the Authoring section, click Assignments, click the ellipsis icon to the right of the assignment you created in the previous task and click Delete assignment.

Not syncing | Policy - Microsoft Azure | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/PolicyMe...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... ALGONQUIN COLLEGE

Home > Policy

Policy | Assignments

Search | Assign policy | Assign initiative | Refresh

Overview | Getting started | Compliance | Remediation | Events | Authoring | Definitions | **Assignments** | Exemptions

Search: Filter by name or ID... | Scope: 3 selected | Definition type: All definition types

Total Assignments: 2 | Initiative Assignments: 1 | Policy Assignments: 1

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative
Inherit the Cost Center tag and its value 000 from the resource group if missing	Azure for Students/CST8912	Policy

Edit assignment | View compliance | View definition | Delete assignment | Duplicate assignment | Create exemption

Give feedback

Not syncing | Policy - Microsoft Azure | Discover - Microsoft Azure | Lab8_Week11 - 24F_CST8912_011

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Policy/PolicyMe...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... ALGONQUIN COLLEGE

Home > Policy

Policy | Assignments

Search | Assign policy | Assign initiative | Refresh

Overview | Getting started | Compliance | Remediation | Events | Authoring | Definitions | **Assignments** | Exemptions

Search: Filter by name or ID... | Scope: 3 selected | Definition type: All definition types

Total Assignments: 1 | Initiative Assignments: 1 | Policy Assignments: 0

Assignment name	Scope	Type
ASC Default (subscription: 469222a6-9aad-44fa-b0b8-12052d695b23)	Azure for Students	Initiative

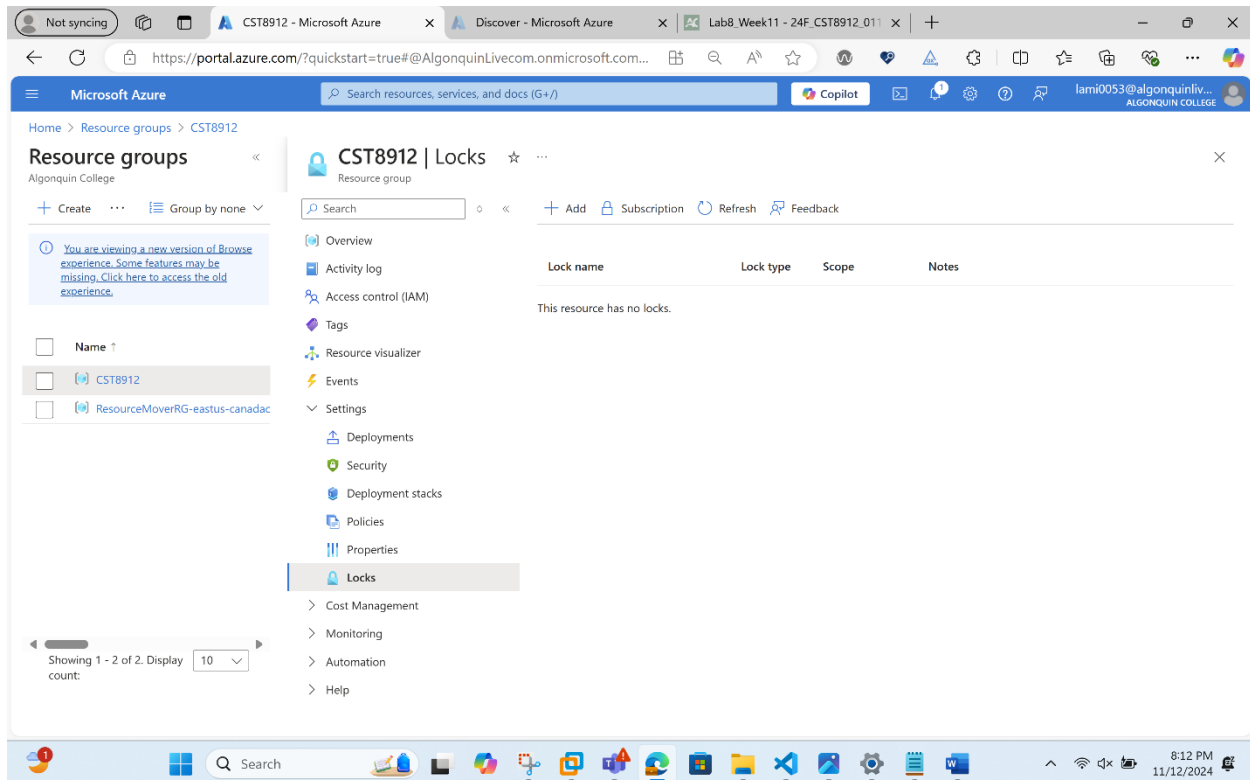
Successfully deleted assignment

Successfully deleted 'Inherit the Cost Center tag and its value 000 from the resource group if missing' from 'Azure for Students/CST8912'. Please note that the change takes around 30 minutes to take effect.

Edit columns

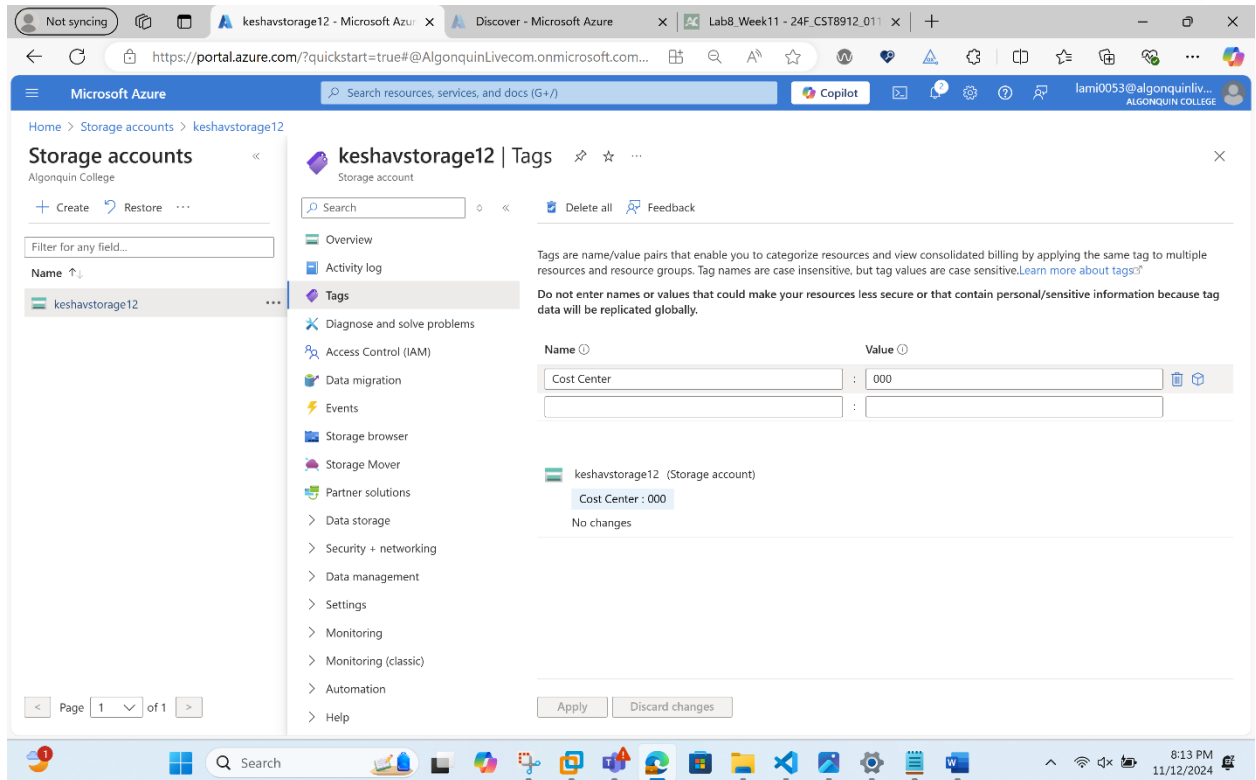
Give feedback

Note: The above assignment can only be deleted once the Locks are deleted as shown below we don't have any locks.



The screenshot displays the Microsoft Azure portal interface. The main content area shows the 'Locks' section for the 'CST8912' resource group. A message states 'This resource has no locks.' The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Deployments, Security, Deployment stacks, Policies, Properties, Locks (selected), Cost Management, Monitoring, Automation, and Help. The top bar shows the user is logged in as 'lami0053@algonquinliv...'.

3. In the portal, search for and select Storage accounts.
4. In the list of storage accounts, select the resource group corresponding to the storage account you created in the last task of this lab. Select Tags and click Delete (Trash can to the right) to the Role:Infra tag and press Apply.



In this step we deleted the Tags.

5. Click Overview and click Delete on the top of the storage account blade. When prompted for the confirmation, in the Delete storage account blade, type the name of the storage account to confirm and click Delete.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Storage accounts' section is active, displaying a list of storage accounts under 'Algonquin College'. The account 'keshavstorage12' is selected. The main pane shows the 'Overview' tab for this account, including details like 'Resource group (move)', 'Location', 'Primary/Secondary Location', 'Subscription (move)', 'Subscription ID', 'Disk state', 'Tags (edit)', and 'Add tags'. On the right, a 'Delete storage account' dialog box is open. It states: 'The following storage account and its contents will be deleted.' Below this, it lists 'Resource to be deleted' as 'keshavstorage12'. A section titled 'Dependent resources to be deleted' explains that data is updated regularly and may take over a day to delete. A table shows no dependent resources. At the bottom of the dialog, there is a confirmation field with the text 'Enter storage account name to confirm deletion *' and the value 'keshavstorage12'. There are 'Delete' and 'Cancel' buttons, and a 'Give feedback' link.

Resource	Number of instances	Total data stored
Containers	-	-
File shares	-	-
Tables	-	-
Queues	-	-

Finally, we deleted the resource group. And this completes the lab.

Not syncing

Delete a resource group - Micros...

Discover - Microsoft Azure

Lab8_Week11 - 24F_CST8912_011

+

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

lami0053@algonquinliv... ALGONQUIN COLLEGE

Home > Resource groups >

Resource groups

Algonquin College

+ Create ... Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

☐ Name ↑

☐ CST8912

☐ ResourceMoverRG-eastus-canadac

Showing 1 - 2 of 2. Display count: 10

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

- Deployments
- Security
- Deployment stacks
- Policies
- Properties
- Locks

Cost Management

Monitoring

Automation

Help

+ Create Manage view

Essentials

Resources Recommendations

Filter for any field...

Type

Showing 0 to 0 of 0 records.

Name ↑

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

CST8912

Dependent resources to be deleted (0)

All dependent resources, including hidden types, are shown

Name	Resource type
------	---------------

Enter resource group name to confirm deletion *

CST8912

Delete Cancel

1

Search

8:14 PM 11/12/2024